Cluster Analysis

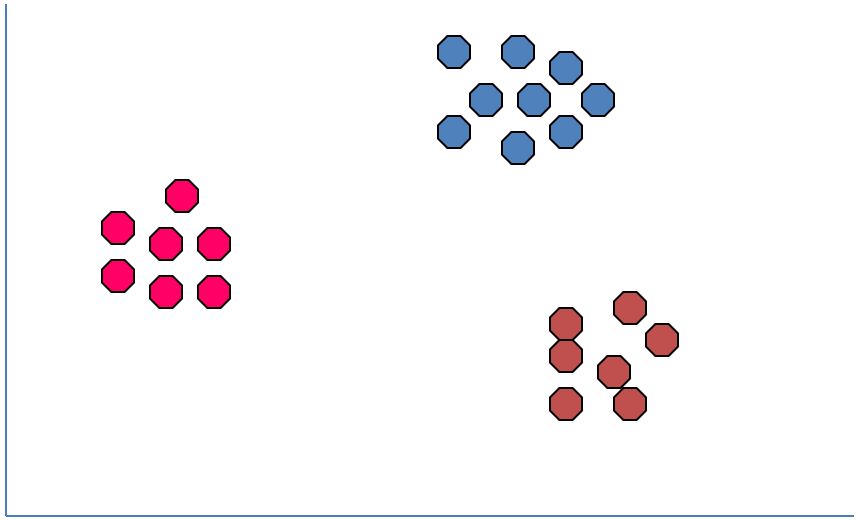
What is Cluster Analysis

- **Cluster analysis** or simply **clustering** is the process of grouping a set of data objects into subsets. Each subset is a cluster, such that objects in a cluster are similar to one another and different from the objects in other clusters.
- **Clustering** is the process of dividing data objects into meaningful group having low intra cluster dissimilarity and high inter cluster dissimilarity.

Cluster Analysis

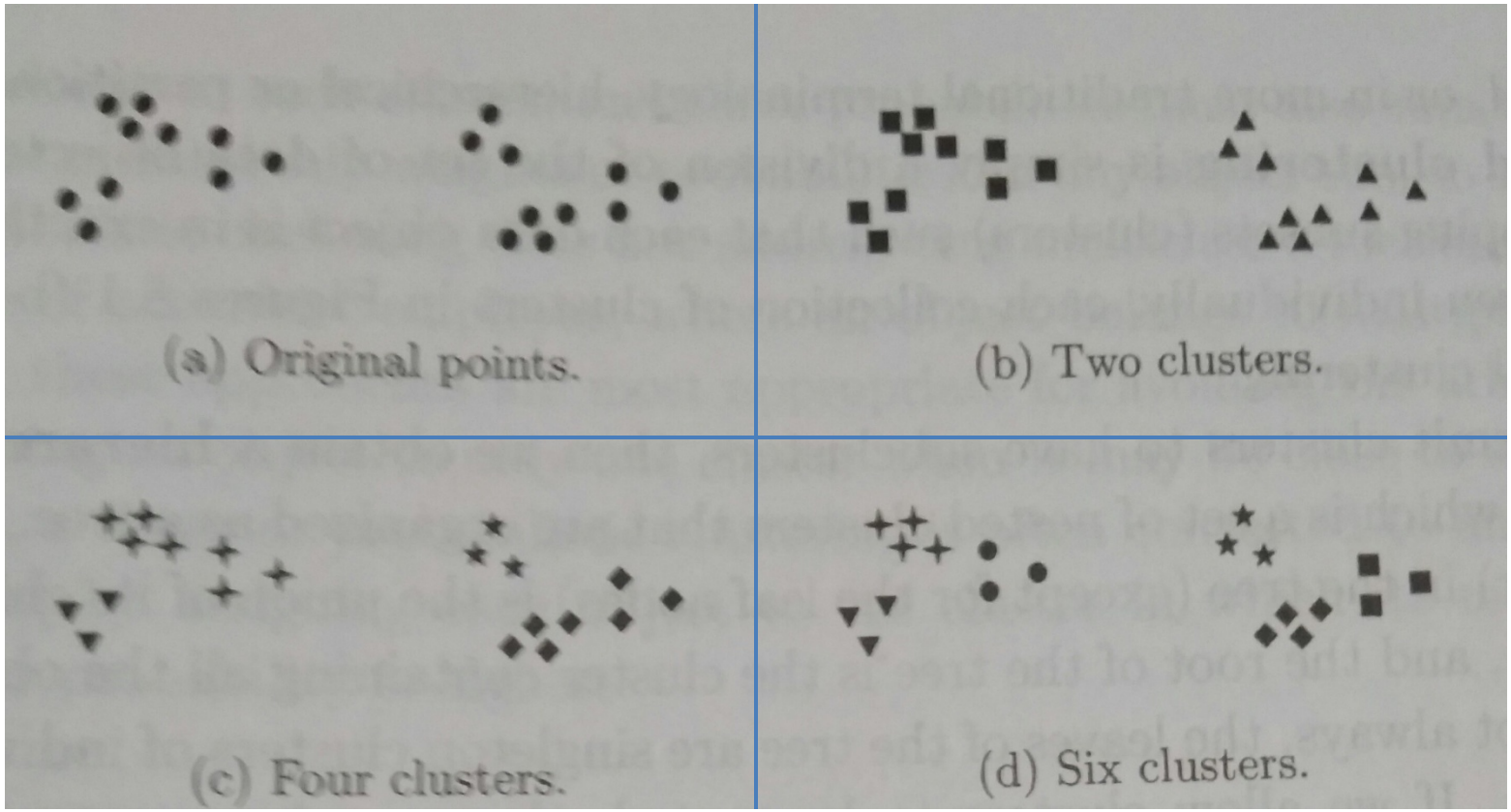
Intra cluster distances are minimized.

Inter cluster distances are maximized.



Cluster Analysis

- Different ways of clustering the same sets of points.



Purpose of Cluster Analysis

- **Clustering for understanding:** Classes, or conceptually meaningful groups of objects that share common characteristcs, play an important role in how people analyze and describe the world.
- **Clustering for utility:** Cluster analysis provides an abstraction from individual data objects to the clusters in which those data objects reside. Some clustering algorithms characterize each cluster in terms of cluster prototype; i.e., a data object that is representative of the other objects in the cluster.

Different types of clusterings and Clusters

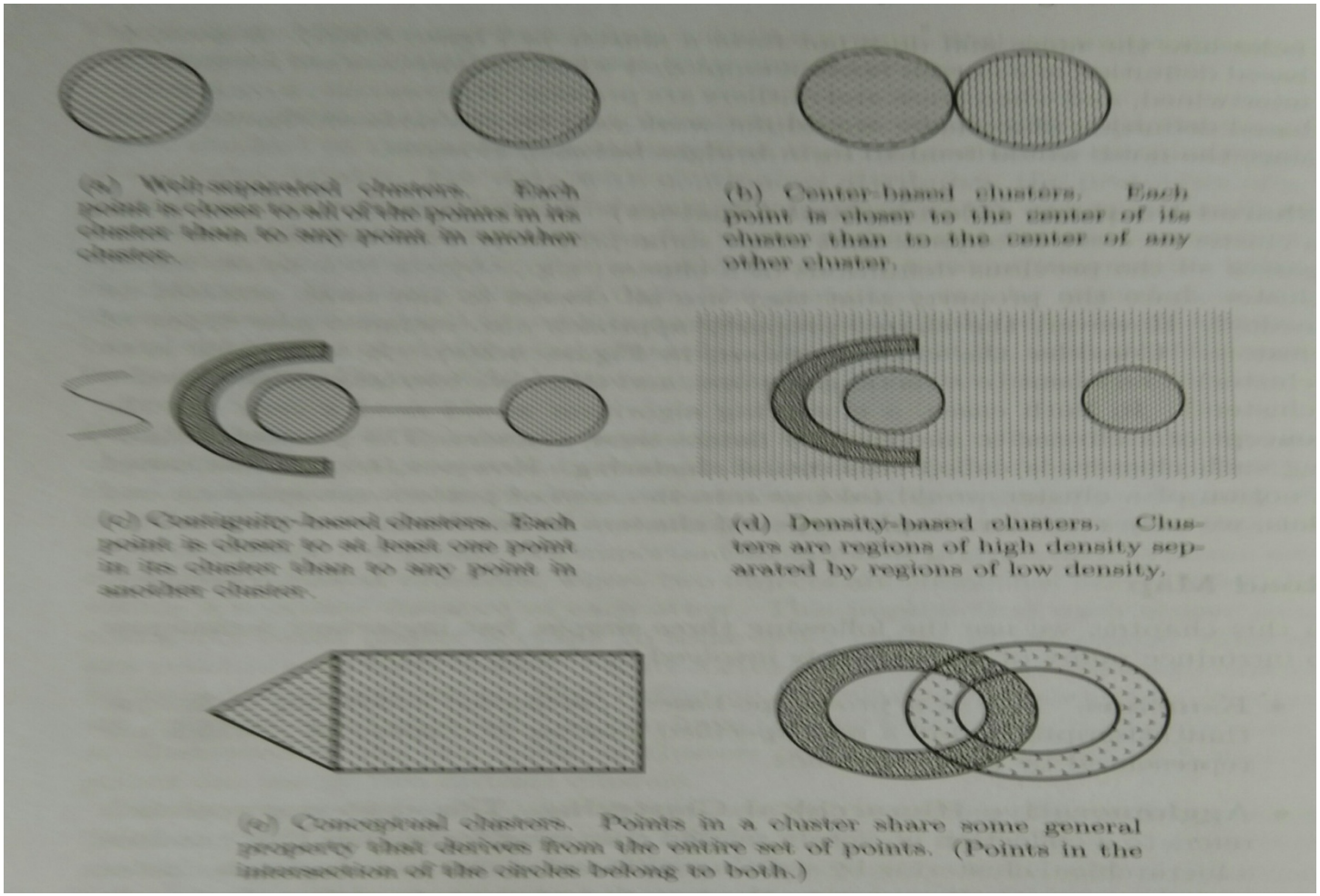
Different types of clusterings

- Hierarchical versus Partitional
- Exclusive versus Overlappinf versus Fuzzy
- Complete versus Partial

Different types of Clusters

- Well-separated
- Prototype-based
- Graph based
- Density-based
- Shared-Property (Conceptual)

Different types of clusters



Requirements for cluster analysis

1. Scalability
2. Ability to deal with different types of clusters
3. Discovery of clusters with arbitrary shape
4. Ability to deal with noisy data
5. Incremental clustering and insensitive to input order
6. Capability of clustering high dimensional data
7. Interpretability and usability
8. Requirements for domain knowledge to determine input parameters

Clustering Techniques

Different types of clustering algorithms:

1. Partitional algorithms
2. Hierarchical algorithms
3. Density-based algorithms
4. Grid-based algorithms
5. Model-based algorithm

clustering methods

Method	General Characteristics
Partitioning methods	•Find mutually exclusive clusters of spherical shape •Distance-based •May use mean or medoid (etc.) to represent cluster center •Effective for small- to medium-size data sets
Hierarchical methods	•Clustering is a hierarchical decomposition (i.e., multiple levels) •Cannot correct erroneous merges or splits •May incorporate other techniques like microclustering or consider object “linkages”
Density-based methods	•Can find arbitrarily shaped clusters •Clusters are dense regions of objects in space that are separated by low-density regions •Cluster density: Each point must have a minimum number of points within its “neighborhood” •May filter out outliers
Grid-based methods	•Use a multiresolution grid data structure •Fast processing time (typically independent of the number of data objects, yet dependent on grid size)

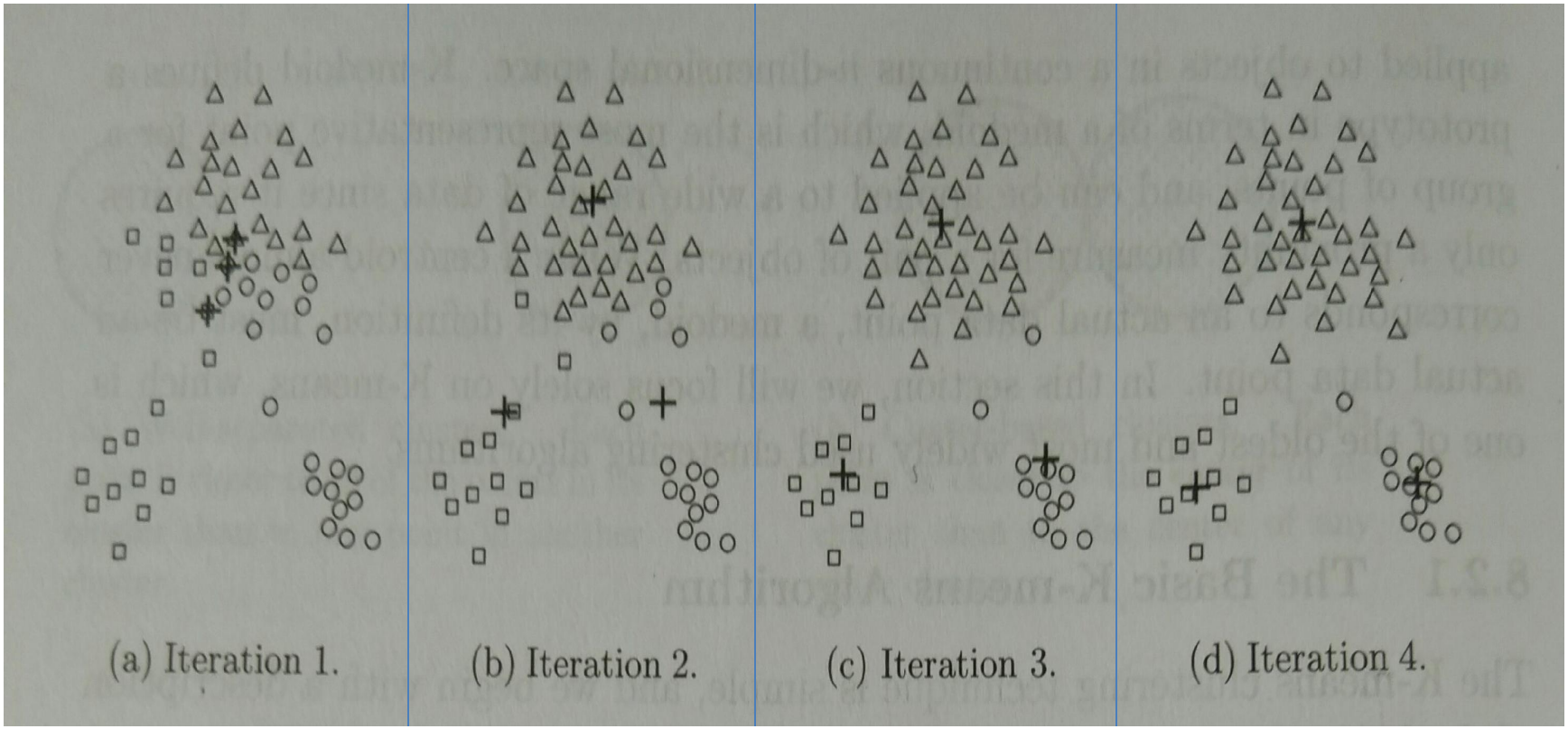
Partitional algorithms

1. K-means algorithm
2. K-medoids algorithm
   i. PAM
   ii. CLARA
   iii. CLARANS

k-Means Clustering algorithm

- Algorithm:
  1. Select k points as initial centroids.
  2. repeat
  3. Form k clusters by assigning each point to its closest centroid.
  4. Recompute the centroid of each cluster.
  5. until centroids do not change

K-means clustering



Objective function

- The goal of the clustering is typically expressed by an objective function that depends on the proximity of points to one another or to the cluster centroids.
- **Sum of the quared error (SSE)** : The objective function of k-means is to minimize the squared distance of each point to its closest centroid. The SSE is formally defined as follows:

$$SSE = \sum_{i=1}^{k} \sum_{x \in C_i} dist(m_i, x)^2$$

- Where m_i is the centroid (mean) of cluster C_i *calculated as:*

$$m_i = \frac{1}{n_i} \sum_{x \in C_i} x$$

- Where n_i is the number of points in C_i.

Problem

Q. Suppose that the data mining task is to cluster points (with (*x, y) representing location)* into three clusters, where the points are:

A1(2, 10), A2(2, 5), A3(8, 4), B1(5,8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4, 9).

The distance function is Euclidean distance. Suppose initially we assign *A1, B1, and C1* as the center of each cluster, respectively. Use the *k-means algorithm to show only*

(a) The three cluster centers after the first round of execution.

(b) The final three clusters.

Problem

1. **Mean1 = *(2, 10)*, Mean2 = *(5, 8)*, Mean3 = *(1, 2)***
2. **For each point starting from 1st point to last point do**:

i. Compute distance of 1st point to each of the three means:

Distance of 1st point A1(2, 10) to Mean1(2, 10)

$d1 = ((2-2)^2+(10-10)^2)^{1/2}=0$

Distance of 1st point A1(2, 10) to Mean2(5, 8)

$d2 =((2-5)^2+(10-8)^2)^{1/2}= \sqrt{(9+4)}= \sqrt{13}$

Distance of 1st point A1(2, 10) to Mean3(1, 2)

$d3 =((2-1)^2+(10-2)^2)^{1/2}= \sqrt{(1+64)}= \sqrt{65}$

The distance d1 is lowest, **1st point is assigned to Mean1.**

ii. Compute distance of 2nd point to each of the three means:

Distance of 2nd point A2(2, 5) to Mean1(2, 10)

$d1 = ((2-2)^2+(5-10)^2)^{1/2}= \sqrt{(0+25)}= \sqrt{25}$

Distance of 2nd point A2(2, 5) to Mean2(5, 8)

$d2 =((2-5)^2+(5-8)^2)^{1/2}= \sqrt{(9+9)}= \sqrt{18}$

Distance of 2nd point A2(2, 5) to Mean3(1, 2)

$d3 =((2-1)^2+(5-2)^2)^{1/2}= \sqrt{(1+9)}= \sqrt{10}$

The distance d13 is lowest, **2nd point is assigned to Mean3.**

Problem

iii. Compute distance of 3rd point to each of the three means:

Distance of 3rd point A3(8, 4) to Mean1(2, 10)

$d1 = ((8-2)^2+(4-10)^2)^{1/2} = \sqrt{(36+36)} = \sqrt{72}$

Distance of 3rd point A3(8, 4) to Mean2(5, 8)

$d2 = ((8-5)^2+(4-8)^2)^{1/2} = \sqrt{(9+16)} = \sqrt{25}$

Distance of 3rd point A3(8, 4) to Mean3(1, 2)

$d3 = ((8-1)^2+(4-2)^2)^{1/2} = \sqrt{(49+4)} = \sqrt{53}$

The distance d2 is lowest, **3rd point is assigned to Mean2.**

iv. Compute distance of 4th point to each of the three means:

Distance of 4th point B1(5, 8) to Mean1(2, 10)

$d1 = ((5-2)^2+(8-10)^2)^{1/2} = \sqrt{(9+4)} = \sqrt{13}$

Distance of 4th point B1(5, 8) to Mean2(5, 8)

$d2 = ((5-5)^2+(8-8)^2)^{1/2} = \sqrt{(0+0)} = 0$

Distance of 4th point B1(5, 8) to Mean3(1, 2)

$d3 = ((5-1)^2+(8-2)^2)^{1/2} = \sqrt{(16+36)} = \sqrt{42}$

The distance d2 is lowest, **4th point is assigned to Mean2.**

Problem

v. Compute distance of 5th point to each of the three means:

Distance of 5th point B2(7, 5) to Mean1(2, 10)

$d1 = ((7-2)^2+(5-10)^2)^{1/2} = \sqrt{(25+25)} = \sqrt{50}$

Distance of 5th point B2(7, 5) to Mean2(5, 8)

$d2 = ((7-5)^2+(5-8)^2)^{1/2} = \sqrt{(4+9)} = \sqrt{13}$

Distance of 5th point B2(7, 5) to Mean3(1, 2)

$d3 = ((7-1)^2+(5-2)^2)^{1/2} = \sqrt{(36+9)} = \sqrt{45}$

The distance d2 is lowest, **5th point is assigned to Mean2.**

vi. Compute distance of 6th point to each of the three means:

Distance of 6th point B3(6, 4) to Mean1(2, 10)

$d1 = ((6-2)^2+(4-10)^2)^{1/2} = \sqrt{(16+36)} = \sqrt{52}$

Distance of 6th point B3(6, 4) to Mean2(5, 8)

$d2 = ((6-5)^2+(4-8)^2)^{1/2} = \sqrt{(1+16)} = \sqrt{17}$

Distance of 6th point B3(6, 4) to Mean3(1, 2)

$d3 = ((6-1)^2+(4-2)^2)^{1/2} = \sqrt{(25+4)} = \sqrt{29}$

The distance d2 is lowest, **6th point is assigned to Mean2.**

Problem

vii. Compute distance of 7th point to each of the three means:

Distance of 5th point C1(1, 2) to Mean1(2, 10)

$d1 = ((1-2)^2+(2-10)^2)^{1/2} = \sqrt{(1+64)} = \sqrt{65}$

Distance of 7th point C1(1, 2) to Mean2(5, 8)

$d2 = ((1-5)^2+(2-8)^2)^{1/2} = \sqrt{(16+36)} = \sqrt{52}$

Distance of 7th point C1(1, 2) to Mean3(1, 2)

$d3 = ((1-1)^2+(2-2)^2)^{1/2} = \sqrt{(0+0)} = 0$

The distance d3 is lowest, **7th point is assigned to Mean3.**

viii. Compute distance of 8th point to each of the three means:

Distance of 8th point C2(4, 9) to Mean1(2, 10)

$d1 = ((4-2)^2+(9-10)^2)^{1/2} = \sqrt{(4+1)} = \sqrt{5}$

Distance of 8th point C2(4, 9) to Mean2(5, 8)

$d2 = ((4-5)^2+(9-8)^2)^{1/2} = \sqrt{(1+1)} = \sqrt{2}$

Distance of 8th point C2(4, 9) to Mean3(1, 2)

$d3 = ((4-1)^2+(9-2)^2)^{1/2} = \sqrt{(9+49)} = \sqrt{50}$

The distance d2 is lowest, **8th point is assigned to Mean2.**

Problem

3. **Recompute the new mean for each Cluster**

   Cluster 1 is formed by the points: A1(2, 10)

   New_Mean1= (2, 10)

   Cluster 2 is formed by points: A3(8, 4), B1(5, 8), B2(7, 5), C2(4, 9)

   New_Mean2=((8, 4)+(5, 8)+(7, 5)+(4, 9))/4=(24, 26)/4=(6, 6.5)

   Cluster 3 is formed by the points: A2(2, 5), C1(1,2)

   New-Mean3 = ((2, 5)+(1, 2))/2=(3, 7)/2=(1.5, 3.5)

- Assign new means to the old means

  Mean1 = New_Mean1

  Mean2 = New_Mean2

  Mean3 = New_Mean3

4. Go to step 2

K-means clustering technique

- **Choosing initial centroids**: When random initialization of centroids is used, different runs of *k*-means typically produce different SSEs. Choosing the proper initial centroids is the key step of the basic *k*-means procedure.
- **Time and Space complexity**:
  - **Space complexity**: $O((n+k)d)$, where n is the number of points, and d is the numer of attributes.
  - **Time complexity**: $O(i.k.n.d)$, where is the number of iterations.

K-means clustering technique

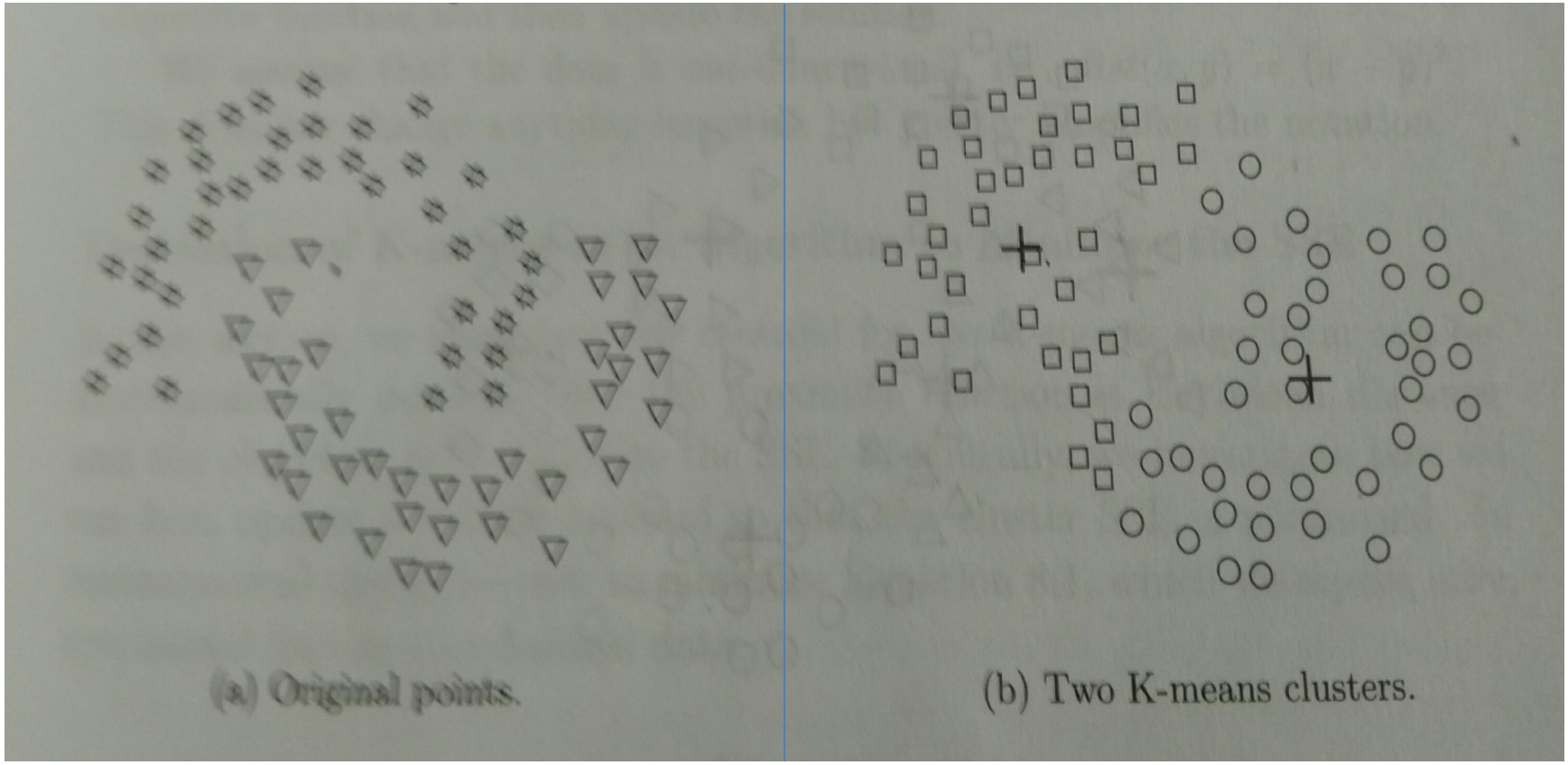
- **K-means: Additional Issues**
- Handling empty clusters
- Outliers
- Reducing SSE with post processing
- Updating centroids incrementally

K-means clustering technique

- **Strength**: K-means is simple and can be used for a wide variety of data types. It is also quite efficient, time complexity is O(n).
- **Weakness**:
  - It can not handle non-globular clusters.
  - It can not handle cluster of different sizes and densities.
  - It is affected by outliers
  - Number of clusters, k is to be predefined.
  - Restricted to data where notion of centroid is defined.
  - Provides locally optimal solution.

K-means clustering technique

K-means with non-globular clusters:

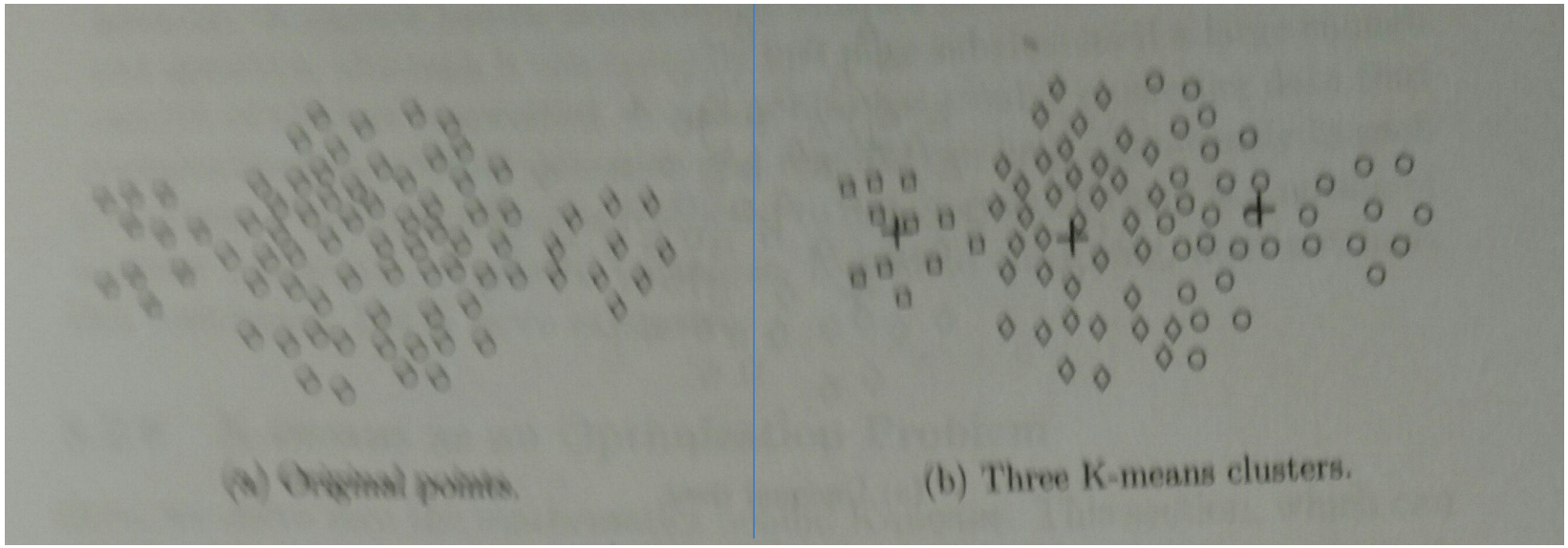


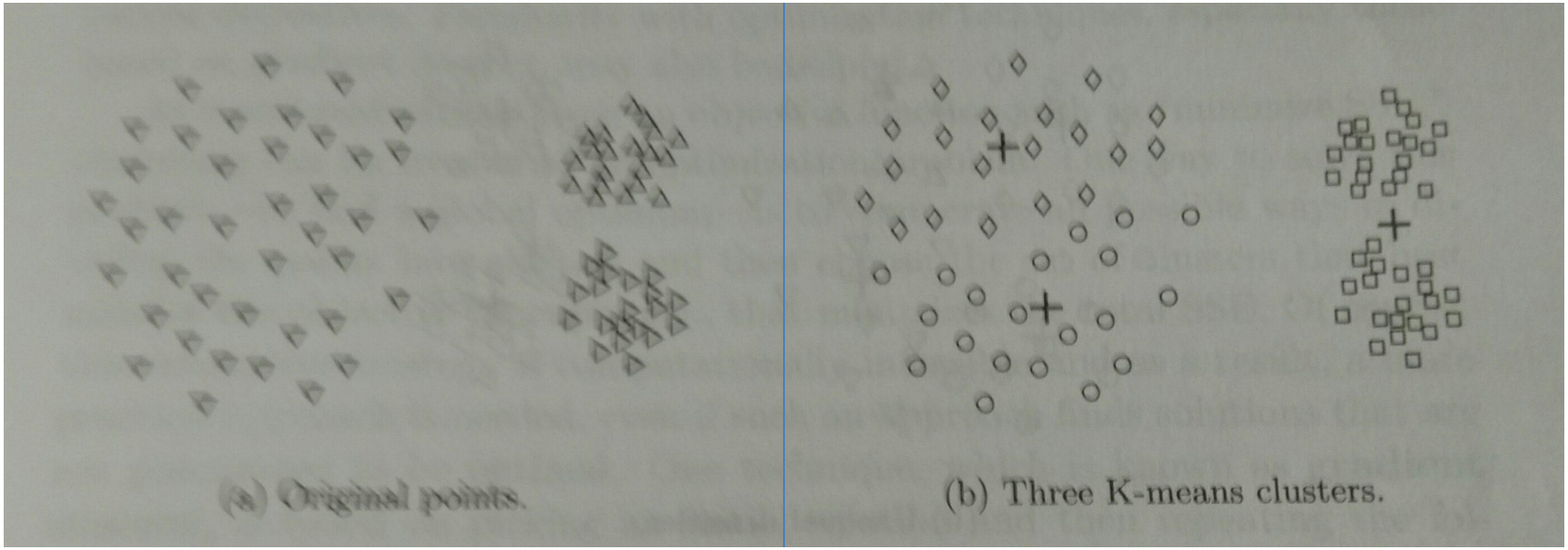
(a) Original points.

(b) Two K-means clusters.

K-means clustering technique

K-means with clusters of different sizes:



K-means clustering technique

K-means with clusters of different densities: